

Research Article

Designing and Manufacturing a Robot for Dry-Cleaning PV Solar Panels

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Solar energy is one of the most important solutions to reduce the concerns of the severe climate change phenomenon. Granted, the main manner to harness solar energy to generate power electricity is implemented through arrays made up of PV solar panels. However, the accumulation of dust on PV surfaces nevertheless remains a serious issue that considerably reduces the efficient conversion of PV panels. Therefore, this research is aimed at automating both monitoring and cleaning of the PV panel's surfaces through the design, manufacture, and operation and evaluating a dry-cleaning robot based on a color monitoring system. The preliminary results demonstrate that the color analysis of the PV panels can distinguish between the density of dust accumulated, where the total color differences between the clean PV panels and both the PV panels with simple, moderate, and intense dust were 43.69, 61.19, and 75.23. This raised the efficiency of the power produced for simple dust panels from 88.03 to 98.91% (one cleaning round), moderate dust panels from 70.72 to 92.96%, and intense dust panels from 39.05 to 62.11% (two cleaning rounds). These preliminary finds illustrate the possibility of using this approach to automatic monitoring the PV panel color and operate the cleaning robot.

1. Introduction

In light of the growing awareness of climate change risks and the increasing greenhouse gas emissions as a result of the negative environmental effects of fossil fuels, for this reason, the global trend focuses to encourage the development of strategies to replace fossil energy generation with clean renewable energy sources. Therefore, generating solar electricity from photovoltaic panels is considered one of the most important global challenges to exploiting clean and renewable energy sources to achieve sustainable development goals and mitigate the effects of climate change [1–3]. In recent decades, there has been a clear growth in the utilization of solar energy, especially in the generation of electrical energy through photovoltaic (PV) panels in most countries of the world, whether developed or developing. Where this growth in solar energy was achieved through the great expansion of the installation of solar panel arrays,

whether on a small scale or the establishment of large stations to serve wide areas and achieve sustainable development [4–6]. Hence, solar energy among all the other clean renewable energy source systems to generate electrical energy has many environmental advantages such as near-zero greenhouse gas emissions, decrease climate change, flexible installation, ease and low cost of maintenance, and being environmentally friendly [7–9]. However, nevertheless, it remains a growing problem represented in the efficient conversion of solar energy systems for electricity generation that is remarkably decreased due to some reasons, such as the ambient overheating temperature, partial shading or cloudy days, and dust accumulation which consider the most important factor that has a negative effect on the PV panels performance [10–19]. In addition to the negative effects of dust accumulation, [20] mentioned that the dust deposition on the PV panels' surfaces leads to other losses such as creating hotspots that may damage PV panels

and reduce the lifespan of the PV panels systems. Generally, dust accumulation leads to a decrease in the solar radiation falling on the PV panels' surfaces and thus causes a fall down in the electrical energy generation of PV panels by a ratio of up to 15%/day, and this low efficiency can reach 30%/day of the total electric power produced especially in intense dusty regions [21–23]. In this regard [24] investigated the performance of a 106 W PV system under weather conditions over one year using a wireless data acquisition and monitoring system. The results found that the exposure of 12 months under different seasons reduced the PV system's efficiency by 24.5, 15.6, 5.14, and 1.95% in summer, winter, postmonsoon, and monsoon, respectively. Also, [25] mentioned that the long periods without rain, such as the summer, dust accumulation can lead to daily losses and can be as high as 20%. Moreover, [26] concluded that soiling losses of PV panels due to dust lead to a significant decrease in the yield of solar energy and result in economic losses. Where it is found that without cleaning this soiling, the output power was decreased by 43% after six months of exposure to an average ambient dust density (0.7 mg/m³). While [27] revealed a decrease of 32% in the efficiency of PV panels in the Algerian Sahara desert after only 6 months without any cleaning process. Similarly, [28] mentioned that if solar panels are not cleaned for one month, the power generation will be reduced by a ratio of 10%. Also, [29] indicated a 20% decrease that was observed in the efficiency of PV panels installed in Freiburg, Germany, over 5 months from the accumulation of dust on glass surfaces. By the same talking, [30] showed that the efficiency of a PV panel can decrease by up to 40% due to the accumulation of residues on the PV panel surface. Furthermore, [10, 31] found that the effect of dust accumulation on the output power of solar panels in real outdoor conditions reached to 50% decrease if the cleaning process is not performed on the panels for some time than six months. On the same approach [32] illustrated that the PV panels' efficiency decreased by 50% after 45 days without cleaning. As a result, there are many recommendations that cleaning operations for PV solar panels should be carried out regularly. As strongly recommended, the photovoltaic panels are cleaned 3 to 4 times a year, especially that weather conditions are not extreme, and this number of cleaning processing should increase during dry periods. [33] showed that the solar panel should be particularly clean to ensure power generation and high efficiency and absorb as much sunlight as possible. These recommendations are implemented through the traditional PV panel cleaning operations which are usually done regularly by human operators, but they have many disadvantages; the most important of which are time-consuming, their accuracy varies from one factor to another, and most importantly, it is very costly where [34] pointed that the cleaning process by human operators generates additional operating costs ranging from 1000 to 4500 €/MW depending on the region. Therefore, there is a necessary need for an automated machine for PV panel cleaning processing to overcome these disadvantages of the traditional cleaning method. Where [35] mentioned that smart systems for cleaning solar panels have the potential to increase energy output, improve life-

time performance, and reduce maintenance costs further—more reducing human intervention. As a consequence, in the last few years, robots for cleaning PV panels have gradually developed and were replaced by traditional cleaning methods implemented by human operators. A robotic device based on programming coding is a systematic and effective method that could be used for solar PV panel stations on large and small scales in cleaning as well as in automatic inspection of panels alongside power consumption and low water utilization when compared to traditional cleaning as mentioned by [36]. Due to the advantages of the robotic device outlined in the previous paragraph, some suggestions have been made for multiple automated systems to solve the problem of dust accumulation on the surfaces of solar panels through robots. Hence, [25] innovated a robot design capable of cleaning and cooling the PV panels, based on images acquired to create a maintaining periodic cleaning system to increase PV efficiency. In the final analysis, the preliminary finds demonstrated the possibility of this approach for the accomplishment of this task. By the same token, [37] designed and developed a robot for consistently cleaning a solar panel by using a rotary brush with water spray integrated with a sun tracker to improve the efficiency of the panel. In a study focused on the design and development of a self-cleaning PV sliding system by [38], the results indicated that the self-cleaning PV sliding system improved the PV efficiency by 18.3%, 13.3%, and 6.4%, respectively, in the summer, winter, and postmonsoon seasons. As well, these results revealed that the energy consumption was very low compared to the amount of energy gained. [39] developed a novel design for the automatic cleaning of solar panels and attached with a water pumping/sprinkling mechanism based on the amount and nature of dust accumulated and found that this system can provide about 30% more energy output when compared to the dust accumulated PV module. Likewise, [40] proposed a similar approach based on a camera mounted directly on the robot, under the control of a mobile application that also collects the data allowing both monitoring and cleaning of both the solar irradiation and the photovoltaic surface, respectively. Also, there was a focus on the concept of a freely moving robot capable of climbing a slope of up to 20°, that could be installed on any solar farm and easy to use and transport [41, 42]). Moreover, [43] presented a novel design for a portable robotic cleaning system for solar panels that can clean and maneuver on the PV panel glass surface at varying angles from horizontal to vertical by using a microcontroller board Arduino Uno R3 to easily control all the robotic parts. It is concluded that an increase in the PV efficiency by 30%–33% is observed on an 8-panel array; hence, it will be of greater use in the solar park. Additionally, Aravind et al. [44] proposed a robot for cleaning PV panels consisting of two subsystems, a robot cleaner for cleaning the PV panel surface, and an automated carrier cart for carrying moves alongside. As a result of what was mentioned above, this research is aimed at monitoring the color of PV panel surfaces and determining the dust density accumulated on the PV panel surfaces through an image processing and analysis color system and then design and manufacture a robot to

TABLE 1: Describing the solar panel specifications.

Model number	STP335-24/Vfw	Power tolerance (positive)	+1.5%
Maximum power (Pmax)	335 W	Current at maximum power (Impp)	8.82A
Voltage at maximum power (Vmpp)	37.7 V	Power tolerance	0/+5 W
Solar cell (mm)	156 × 156	Number of cells	60 (6 × 10)
Dimensions (mm)	1960 × 992 × 40mm	Weight (kgs)	22.1
Panel efficiency	17.2	Operating temperature	−40 °C – +85 °C
Short circuit current (Isc)	9.51A	Open circuit voltage (Voc)	46.5 V



FIGURE 1: Demonstrates the installation of the PV solar panels.

clean PV panels and raise their efficiency, based on color analysis data for the PV panels surfaces with different dust densities compared to the standard color of clean PV panels.

2. Material and Methods

In this section, a preliminary design and manufacture of an automated robot for cleaning PV panels will be presented to suit the local climatic conditions in Egypt. This technological solution based on a robot attached to a machine vision camera to clean the solar panels' surface is aimed at overcoming both the high energy losses and the low efficiency of cleaning conventional systems. All the experiments that were conducted to manufacture this system were carried out in Fab Lab Egypt, Villa 35-100 st.-Near Al Horia Square, Maadi, Cairo and the library of the American Center Cairo (ACC) Digital Library Service-U.S. Embassy in Egypt, Tawfik Diab St, Qasr Ad Dobarah, Qasr El Nil, Cairo Governorate. This experiment was carried out in three successive steps, identifying solar PV panel samples, then monitoring, measuring, and analyzing the color of clean PV panel samples (standard color) and PV panel samples with different dust densities. Finally, the design, manufacture, operation, and evaluation of an automated robot for cleaning solar panels consist of two main parts in hardware and software as follows.

2.1. Solar PV Panel Samples. Two solar PV panels are connected in series, the capacity of each panel is 335 W, and their total is 670 W, to test, operate, and evaluate the proposed cleaning robot. The specifications of the solar PV panel used are shown in Table 1. As well, the panels were installed on a galvanized iron chassis with dimensions 250 × 130 cm (length and height) and a thickness of 3 mm in the southern direction at a tilt angle of 30° as shown in Figure 1.

2.2. Solar PV Panel Color Analysis. Several solar PV panel samples varied in dust density (clean, simple, moderate, and intense dust density), as shown in Figure 2, were used to analyze and measure color parameters upon solar panel surfaces at these different dust densities.

An image processing system was applied to color analysis where the image capture system consists of a machine vision camera (Model SXY-I30 equipped with lens 2/3" megapixel) in a vertical position facing solar cells. It is installed on a stand on a height of one meter from the ground in front of these panels. The camera is connected to a laptop Intel core i5-3320M, 3.30 GHz, 8GB physical memory, and Microsoft Windows 10. The solar cells were imaged with a resolution of 480,000 pixels, and two images were acquired for each solar cell in the daylight for subsequent color analysis. A median filter was used to process and remove noise from whole images during color feature extraction using the MATLAB (R2013b) program.

$$C^* = \sqrt{a^{*2} + b^{*2}} \dots, \quad (1)$$

$$h^* = \tan^{-1} \left(\frac{b^*}{a^*} \right) \dots, \quad (2)$$

$$\Delta L^* = L_1^* - L_2^* \dots, \quad (3)$$

$$\Delta a^* = a_1^* - a_2^* \dots, \quad (4)$$

$$\Delta b^* = b_1^* - b_2^* \dots, \quad (5)$$

$$\Delta E_{ab}^* = \sqrt{\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2}} \dots, \quad (6)$$

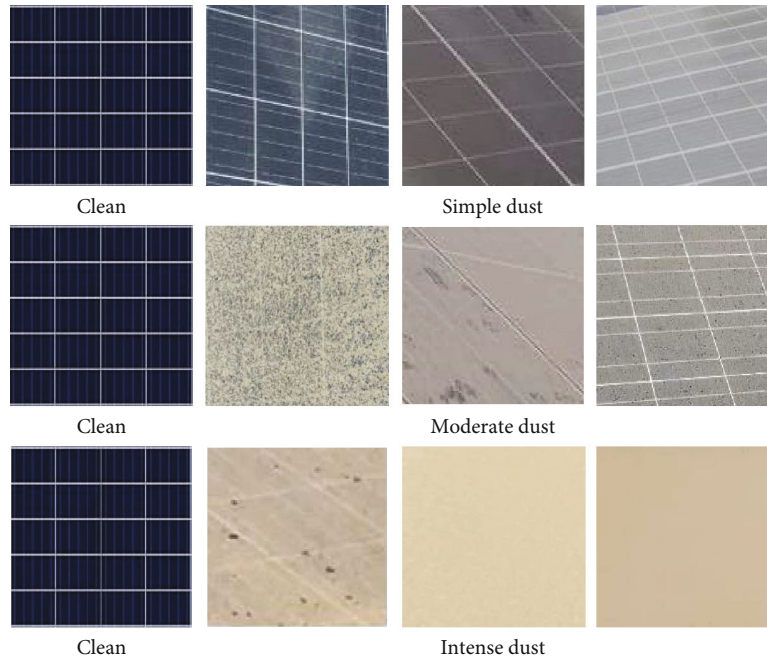


FIGURE 2: Shows images of different dust densities on the solar cell surface.

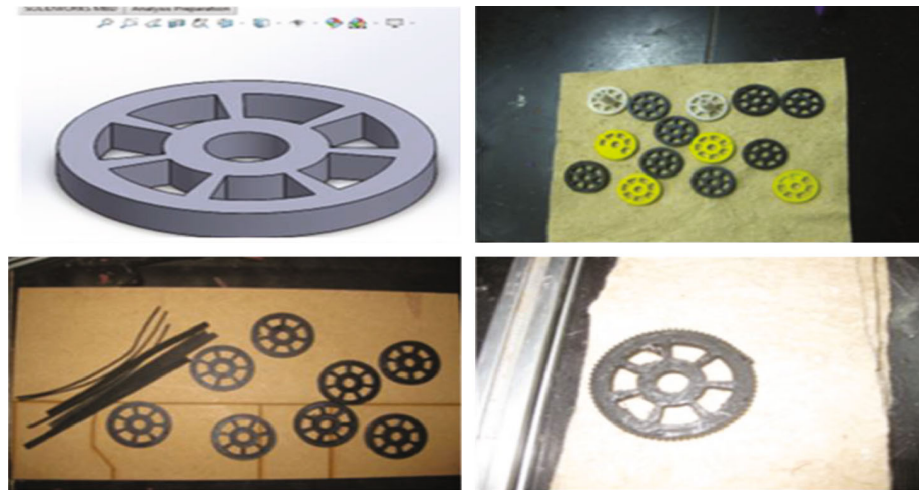


FIGURE 3: Shows the design, 3D printing, and manufacturing of the movement wheel.

where L^* refers to the lightness component from 0 black to 100 white, a^* and b^* are the chromaticity coordinates represented in the x color axis from red (+) to green (−), and the y color axis indicates yellow (+) to blue (−), respectively. While L_1^* , a_1^* , and b_1^* represented the color parameters of clean PV panel sample (standard color sample), and the L_2^* , a_2^* , and b_2^* are the color parameters of different dust densities PV panel samples (simple, moderate, and intense dust density). Also, ΔL^* , Δa^* , Δb^* , and ΔE_{ab}^* are expressing for differences in both lightness/darkness values (+ = lighter and − = darker), the difference on the red/green axis (+ = redder and − = greener), the difference on yellow/blue axis (+ = yellower and − = bluer), and difference on total color, respectively [45, 46].

2.3. Robot Hardware. The hardware of the solar panel cleaning robot is composed of a main frame, wheels, cleaning head, and DC motors that enable the cleaning head to move along the panels to clean the whole surface. 3D printer (Model: i3 MK3, Prusa, Czech) with a working volume (of $25 \times 21 \times 21$ cm) and laser cutters powered 90 watts (Model: MD 3050D, Morn, China) at working area: 30×50 cm were used to implement manufacturing of this parts.

- (i) A set of wheels were designed on the SolidWorks program with a diameter of 45 mm and thickness of 6 mm (Figure 3) to provide the main frame and cleaning head with horizontal and vertical movement to enable it to carry out the cleaning process.

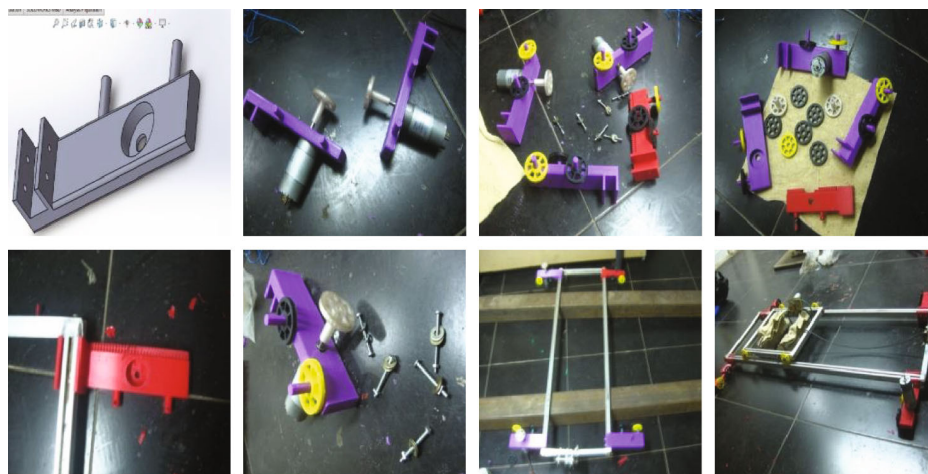


FIGURE 4: Describes the design, 3D printing, and manufacturing of the whole main frame.

Wheel edges are covered with a leather layer to minimize friction and ease of movement during the motion position

- (ii) The main frame of the cleaning robot was designed using lightweight material such as aluminum bars with dimensions 130×40 cm, to provide the path of the cleaning head carriage and installed on eight wheels to support horizontal movement between cells through two DC motors 12 V, 800 mA, and 300 rpm. To facilitate the horizontal movement of the robot on the solar panels' surface, the main frame attached to the wheels responsible for the horizontal displacement to move between the solar panels has been carried out and fixed into four main parts. The shape of each main part was designed to contain a cavity with a diameter of 45 mm including a hole with a diameter of 20 mm to pass the motor shaft in addition to three holes to install the wheels for horizontal movement as shown in Figure 4
- (iii) *Cleaning head trolley.* The cleaning head moves across the solar panels through a motorized trolley which travels along the top to bottom with the help of aluminum rods using a track belt within the edges of the solar panel to clean the panels as shown in Figure 5. The chassis of this cleaning head trolley is made up of aluminum rails with dimensions 40×40 cm and is mounted on eight wheels and two wheels in each corner with a diameter of 40 mm, to enable the movement up and down and vice versa along the length of the solar cell through a DC motor 12 V, 800 mA, and 300 rpm

Also, it consists of a head that has two brushes in a cylindrical hollow shape of a length of 40 cm with an outer diameter of 10 cm weighing 150 g. These brushes are covered with a type of nonstick cloth (double pieces for each pillar of the cylindrical brush) and are characterized by their ability to clean. Finally, the brushes are rotated with the help of a DC motor at 12 V, 800 mA, and 300 rpm which gives a tor-

que of 1.3 Kgf. cm. The wheels are installed in the frame of the cleaning head trolley, using a shaft with a diameter of 10 mm that contains a hole for fixing the wheels with a screw to prevent them from deviating from the movement track. Furthermore, the winch part which consists of a wire connected to a motor installed in the control unit to move the cleaning head trolley from top to bottom and back is shown in Figure 6. When the clean operation starts, the robot immediately begins to disassemble the wire, and the cleaning head trolley goes down with the cleaning brushes running until it reaches the solar panel bottom and touches the limit switch to reflect the winch motor movement to pull the cleaning head trolley up.

2.4. Robot Software. The software of the cleaning robot is programmed with a microcontroller that controls the hardware operations and its vertical and horizontal movements from the PV panel to the other panel of the cleaning robot. This control unit is composed of Arduino Uno (microchip ATmega328P), two limit switches, two resistance 2200 ohm, two motor drives, an Arduino Bluetooth module, a mobile application (Bluetooth RC), and a battery as shown in Figure 7. Arduino Uno as an open-source microcontroller board based on the microchip ATmega328P microcontroller is used to control four motors. Initially, controlling the first two motors provides the movements to both brushes and the trolley. Where the main idea of controlling these two motors is to turn on the 1st motor to move the cleaning trolley from the bottom to the top, at the same time, it turns off the 2nd motor responsible for the rotational movement of the cleaning brushes until the trolley reaches the top. Hence, when the robot reaches the upper edge of the solar cell, the movement of the cleaning trolley is separated through turning off the 1st motor, and the 2nd motor starts to operate the cleaning brushes and moves down under the inertia of the robot. Then comes the role of the limit switch keys or delay work where the time taken by the robot for one cleaning cycle is calculated, and then, the control unit operates the motors responsible for the horizontal movement along the solar cells. It controls the horizontal displacement that

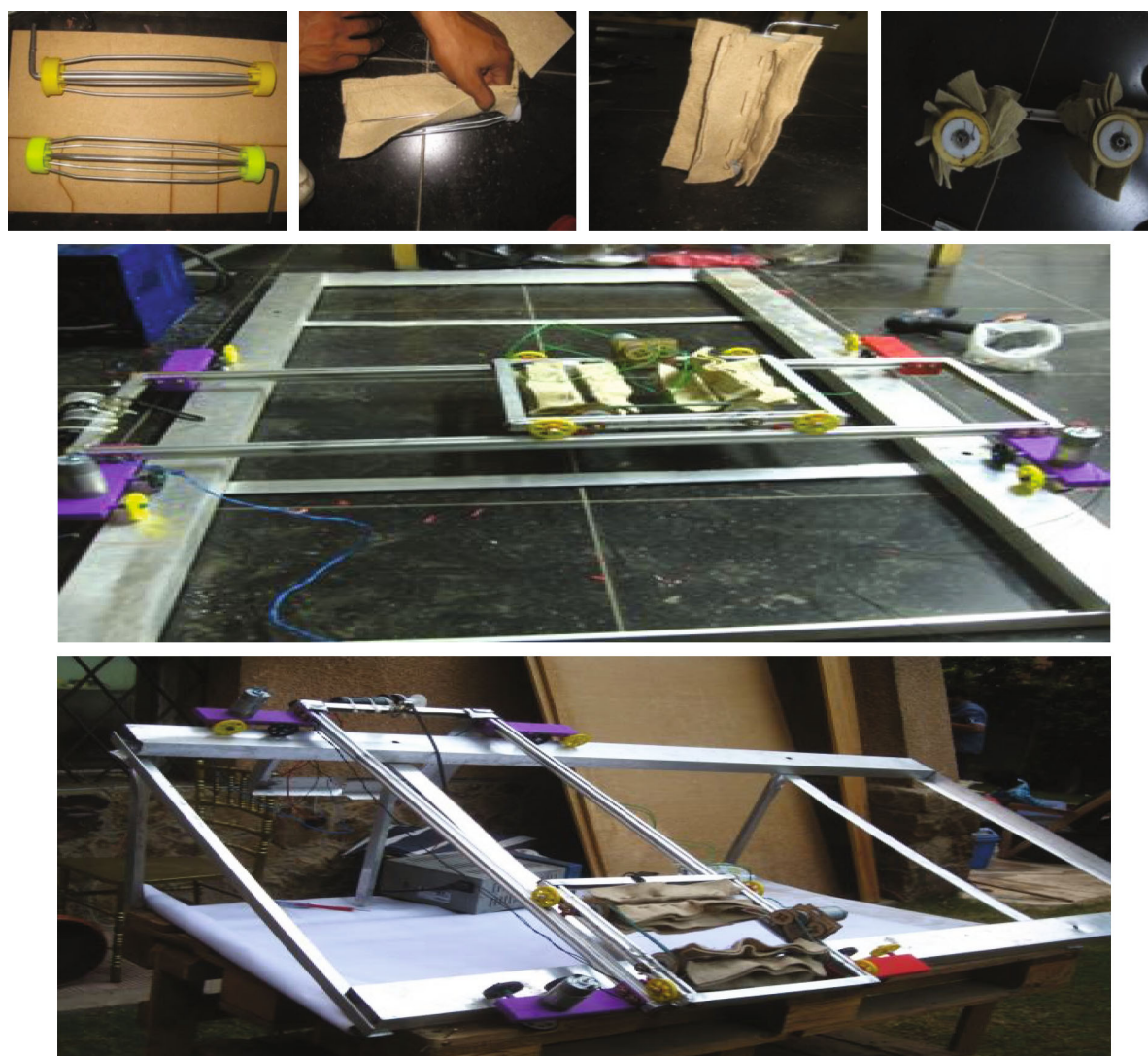


FIGURE 5: Shows a panorama of the manufacturing steps of the cleaning head trolley.

enables the mainframe to move horizontally after the cleaning head has gone up and down (clean cycle), to a distance of 35 cm to travel to another solar cell to complete the cleaning process by controlling the main frame motors (3rd and 4th motors) as shown in Figure 8.

Moreover, motor drives, an H bridge motor drive (RKI 1004) (U1 and U2), are an electronic circuit that switches the polarity of a voltage applied to a load, for reversing the current, and were used to allow the DC motors to run forwards or backward. All components of the circuit derive the energy needed to run them through a lead acid battery (battery 12 V 2.3A hour).

An independent PV panel with small dimensions of 35 * 45 cm and a maximum output power of 20 W is installed at the bottom of the robot to supply the battery with electric charge. It is controlled through a charge controller model MPPT solar battery charger with a current 40/50/60/100 A, USB LCD, and voltage 12/24 V. Finally, mobile is used to control cleaning robot movements through uses of the Bluetooth module. The programming code was generated

according to the type and specifications of the used solar PV panels mentioned in Table 1. All the programming codes for the running and controlling of all operations of the robot cleaning system were written on an Arduino Uno board (Figure 9).

The cleaning processing of PV panels by the designed robot consists of three steps: start to run the system, then action to move the trolley down, and move the brushes to clean the PV panel surface in the meantime. Once the trolley reaches down and touches the limit switch, hence changes the movement of the cleaning trolley upwards while stopping the movement of the brushes until it reached the top of the solar panel surface. Finally, in the last step, the main frame is moved horizontally for a distance of 35 cm to conduct a new cleaning process for a new place on the surface of the solar panel as shown in the flowchart of the sequence steps of the cleaning programming code (Figure 10).

2.5. Electric Power Calculation. The power produced by the solar cells was calculated by measuring the electric current.

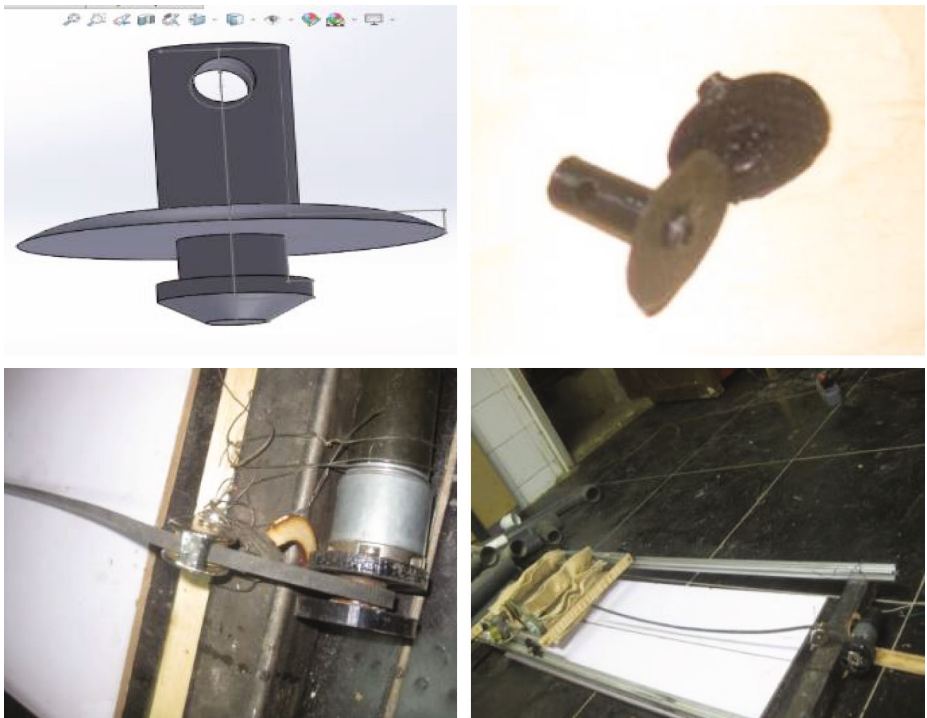


FIGURE 6: Shows a fixing wheels shaft and the winch.

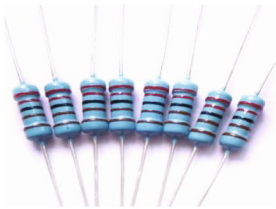
Arduino uno



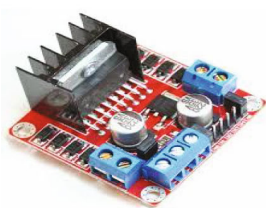
Limit switch



Resistance



Motor drive



Lead acid battery



Arduino bluetooth module



Mobile application (bluetooth RC)

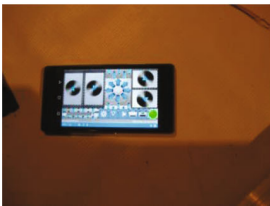


FIGURE 7: Clarify control unit components.

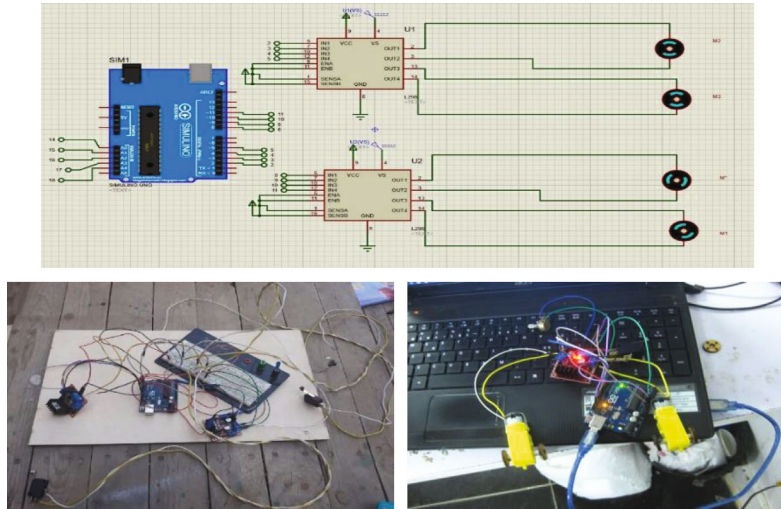


FIGURE 8: Illustrates Arduino board control.



FIGURE 9: Shows the written programming code.

The current was measured by using a hand-held 3 5/6 bit digital multimeter (UNI-T UT89X) device with true RMS sine wave measurement. It can measure Max. 1000 VDC/AC voltage and Max. 20 ADC/AC. As well, it is equipped with audible and visible alarms, a flashlight, and auto back-light functions. Then, the output electrical power was calculated by current intensity through the following equation:

$$P = I \times V \dots, \quad (7)$$

where p is power in watts, V is the voltage in volts, and I is the current in amperes (DC).

3. Results and Discussions

3.1. Color Analysis of Different Solar PV Panels. Primary color parameters red (R), green (G), and blue (B) of clean PV panels (standard color) compared to PV panels with different dusty densities such as simple, moderate, and intense dusty were analyzed and presented in Figure 11. As a result of the dark blue color of the clean solar panels (standard color), it was observed that the lowest values for the primary color model (RGB color model) compared to different dusty panels were 20.51 ± 19.73 , 31.86 ± 19.97 , and 65.77 ± 19.52 for R, G, and B, respectively. Conversely, the panels with

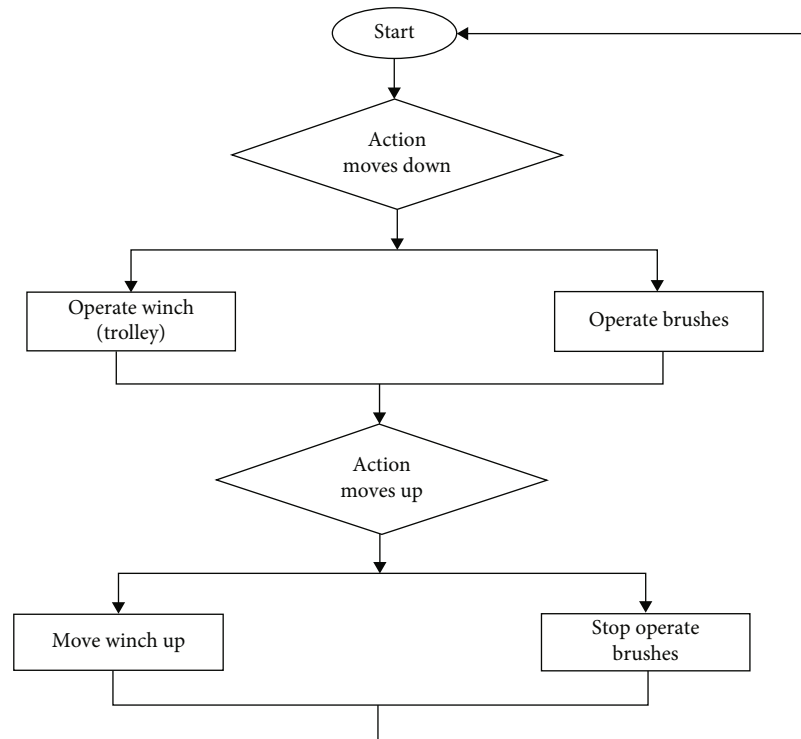


FIGURE 10: Shows the flowchart of the sequence cleaning programming code steps.

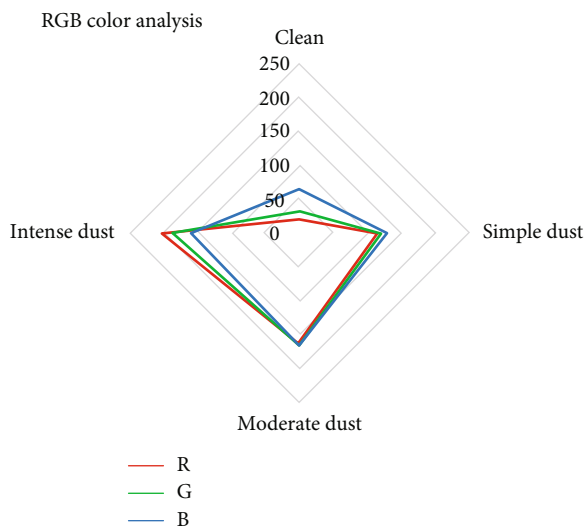


FIGURE 11: Describes the RGB color values of different dust densities on the PV panel surfaces.

intense dust achieved the highest values for both R (204.79 ± 11.87) and G (188.33 ± 12.40) while the B value (161.03 ± 7.19) comes in the second ranking after the panels with moderate dust with a B value of 165.88 ± 23.89 .

The solar panels with simple and moderate dust densities occupied the middle zone, between the lowest value in clean panels and the highest value in panels with intense dust, respectively. Subsequently, lab color parameter results obtained for clean PV panels, and PV panels with different dusty densities (simple, moderate, and intense dust) showed

that the lightness (L^* value) of clean panels ranged from 5.90 to 66.57 with Avg + Std 12.86 ± 8.59 and were less than different dusty densities PV panels (simple, moderate, and intense dust) with average values 51 ± 14.38 , 68.05 ± 5.59 and 77.04 ± 4.33 , for simple, moderate, and intense dust PV panels, respectively. This is due to the dark blue color of the clean panels compared to the dusty color, one of the shades of pale yellow. While the values of a^* (red/green) and b^* (yellow/blue) color parameters of clean PV panels achieved the highest positive value (8.3 ± 1.73) and the lowest negative value (-23.472 ± 1.57) for a^* and b^* , respectively. In contrast, a^* and b^* color parameters of different dusty densities of PV panels (simple, moderate, and intense dust) achieved average values as shown in Figure 12. Moreover, hue angle (h^*) and chroma (C^*) are other significant color factors that were calculated from RGB and lab color models, where the standard values of h^* and C^* for clean PV panels were value averages of 289.23 ± 14.5 and 24.93 ± 1.5 , respectively. All shades of blue are concentrated on the color wheel in the fourth quadrant, starting from angle 270 to 360. On the other hand, the solar panels with intense dust density recorded the lowest value of the color angle in the range from 74.90 to 112.32 with an average of 83.81 ± 4.13 , which was noted at chroma (C^*) value with an average of 15.41 ± 1.5 . The color hue angle (h^*) also changed significantly in each of the solar panels with simple and moderate dust densities, where it was observed in the third and second quadrants of the color wheel with average angles of 212.05 ± 10.5 and 136.72 ± 10.5 with chroma values (C^*) lower than that of clean PV panels, with averages of 5.7 ± 1.5 and 4.46 ± 1.5 for simple and moderate dust densities' samples, respectively, as shown in Figure 13. These lower

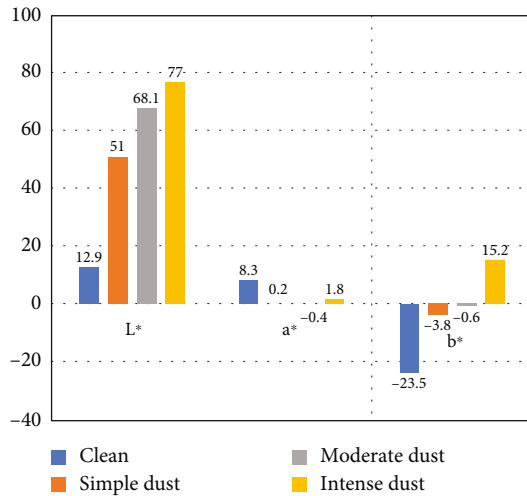


FIGURE 12: Demonstrates the intensity of the L^* , a^* , and b^* color model for PV panels with different dust densities.

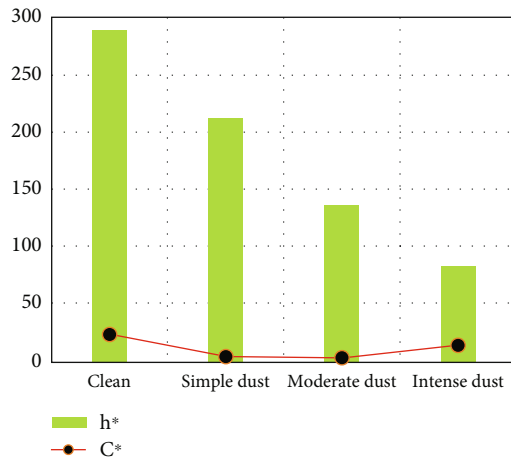


FIGURE 13: Hue angle (h^*) and chroma (C^*) changes in different dust densities of solar panels.

chroma (C^*) values compared to the chroma of the blue dark color for the clean PV panels indicate an increase in the dust density on the surface of the PV panels.

Finally, the total color differences of the color parameters of clean and different dusty densities of PV panels are tabulated in Table 2. It is clear from Table 2 that the values of the total color difference (ΔE^*_{ab}) between the standard color of the clean PV panels and the dust color of the different dusty densities of the PV panels showed a highly significant variation in the color factor.

The higher the dust density, the greater the ΔE^*_{ab} with values ranging from 43.69, 61.19, and 75.23 for simple, moderate, and intense dust, respectively, compared to the zero value of the clean PV panels (standard color). This high difference in the ΔE^*_{ab} is a result of each of the total changes in lightness (ΔL^*), which ranged from -38.14, -56.69, and -64.18. Also, as a result of the total change in the values of both Δa^* (+redness to - greenness) with values 8.10, 8.28, and 6.52 and Δb^* (+yellowness to - blueness) with values -19.70, -21.49, and -38.72.

TABLE 2: Color parameter differences between clean (standard color) and dusted PV panels.

PV panel samples	ΔL^*	Δa^*	Δb^*	Δh^*	ΔC^*	ΔE^*_{ab}
Clean	0	0	0	0	0	0
Simple dust	-38.14	8.10	-19.70	77.18	19.23	43.69
Moderate dust	-56.69	8.28	-21.49	152.51	20.47	61.19
Intense dust	-64.18	6.52	-38.72	205.42	9.52	75.23

Furthermore, the color hue angle (h^*) showed a significant difference between the h of the standard color angle and the h of the colors with different dusty densities, as well as the total change in chroma (ΔC^*) as shown in Table 2. These significant differences in the primary color model values, lab color model, and calculated color parameters hue angle (h^*) and chroma (C^*) between clean solar panels (standard color) and others with different dusty densities may enable us to monitor the cleanliness of the solar panels and take the decision to carry out the cleaning process automatically. Thus, it can specify one or more cleaning round for the same PV panel surface according to the value of the total change in hue angle (Δh^*) or the total color difference (ΔE^*_{ab}), without referring to traditional methods.

3.2. Solar Radiation and Power Produced before and after the Cleaning Process. The measurements of solar radiation intensity (W/m^2) and the output power of the different PV panels (clean, simple, moderate, and intense dust PV panels) were carried out at Cairo (30.05°N and 31.17°E) by using a solar radiation meter (Model PCE-SPM 1) and multimeter (Model UNI-T UT89X). The solar radiation intensity (W/m^2) was measured every hour throughout the daylight during the experiment period, and the averages of this intensity were presented in Figure 14.

It is noticed from Figure 14 that the intensity of the solar radiation evolves with the gradation of the daylight hours from the morning, noon, and afternoon, where the solar radiation intensity increases gradually from the morning period by a value of 565 W/m^2 until it reaches the beginning of the afternoon, hours 12-13 with value 790 W/m^2 , and then gradually decreases until its value reaches 557 W/m^2 at time hours from 15 to 16. This is due to the beginning of the sunset phenomenon. These results of the solar radiation intensity led to similar results in the production of power from different solar panels as shown in Figure 15. The highest value of the intensity of solar radiation corresponds to the high value of the power produced from the solar panels at clean solar panels and time 12-13 hours with a value of 603 W. While the highest value of output power from the solar panels was 607 W at a value of the intensity of solar radiation 788 W/m^2 slightly lower than the maximum value of the intensity of solar radiation. This is due to the beginning of the slight change in the vertical position of the sun's rays to tilted and thus leads to a slight decrease in temperature, which increases the efficiency of work solar panels in production capacity.

While for clean solar panels, the lowest value of the produced power was 400 W attached with the lowest intensity of

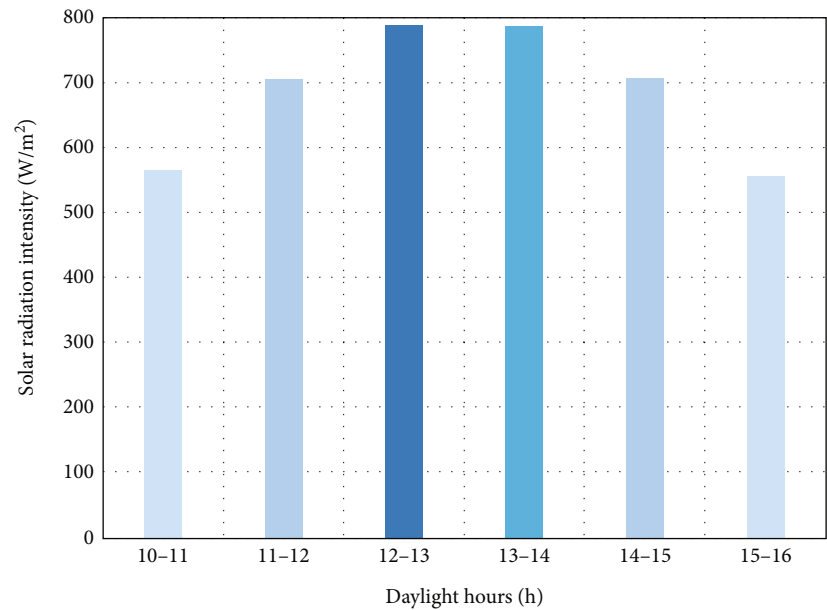


FIGURE 14: Describes solar radiation intensity evolution over the daylight hours.

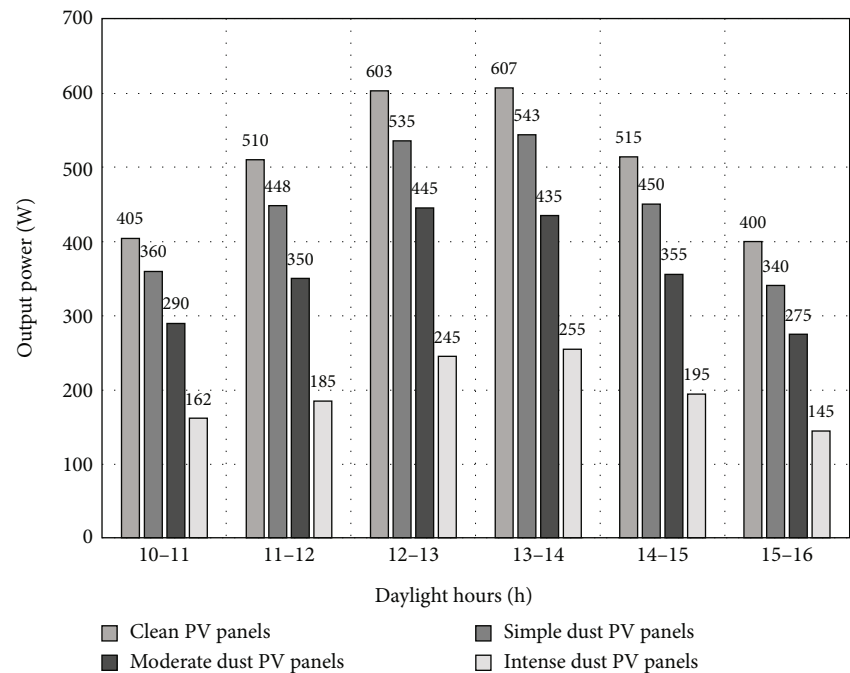


FIGURE 15: Highlights the power produced by the surfaces of clean and different dust-density PV panels.

solar radiation 557 W/m^2 at the time 15 to 16 hours. On the other hand, as for the solar panels with different dust densities, there was a clear and increasing decrease from panels with simple dust to the maximum deterioration in the power production of panels with intense dust. As for the simple dust PV panels, a decrease in productive power was recorded from 11% to 21.8%. This decrease in power production increased dramatically as the dust density increased on the surfaces of the PV panels from simple dust to both moderate and intense dust. The percentage of decrease for solar panels with medium dust density ranged from 26.20 to 31.25%.

While the highest deterioration of the percentages of decrease in the power produced from the PV panels with intense dust reached 57.9 to 63.75%. As a result of the great variance in the results of the power produced by the different solar panels in the degree of surface cleanliness, the cleaning robot was operated to clean the dust-density gradient solar panels, where the time required to carry out one cleaning round was from 10 seconds at a speed of $300 \text{ rpm} = \sim (0.7 \text{ m/s})$. As a consequence, Figure 16 shows the significant difference in the power produced after performing the cleaning process for different panel surfaces in the density of dust

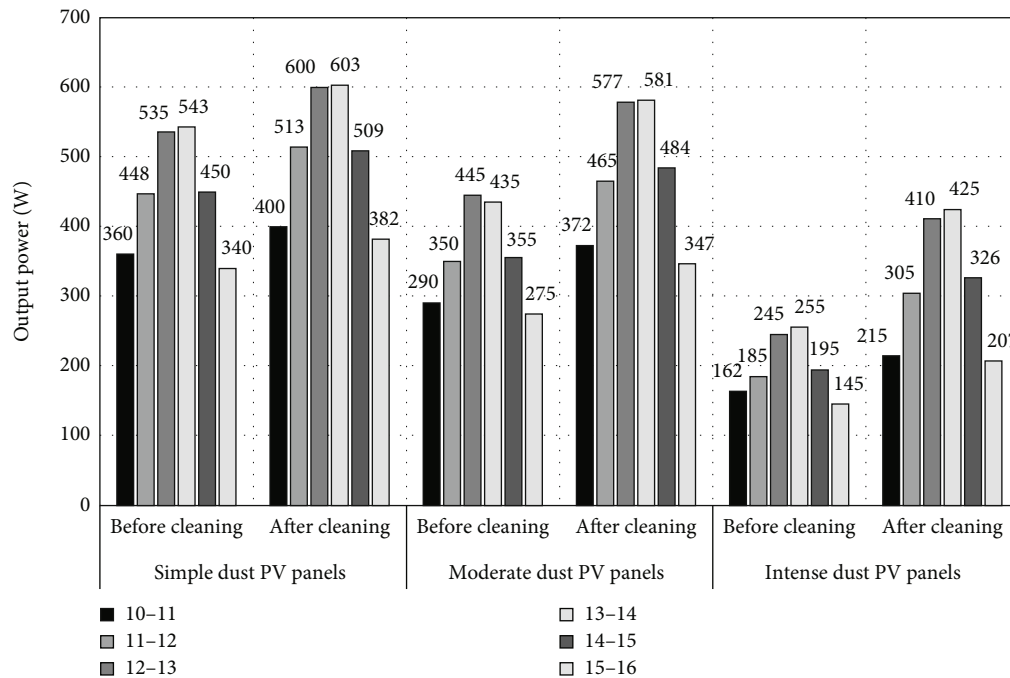


FIGURE 16: Illustrates the variance between the power produced by PV panels before and after the cleaning process.

TABLE 3: Shows the differences in power produced and color coefficients and cleaning efficiency.

PV panel samples		Avg.±SD output power (W)/day	CV	SE	Δ Output power (W)	ΔE* _{ab}	Number cleaning rounds	Time (s)	Efficiency (%)	Δ Efficiency (%)
Standard	Clean PV panels	506.67 ± 90.71	0.18	1.65	0	0	0	0	100	0
Before cleaning	Simple dust PV panels	446.00 ± 84.83	0.19	1.64	60.67	43.69	0	0	88.03	11.97
	Moderate dust PV panels	358.33 ± 70.83	0.20	1.73	148.34	61.19	0	0	70.72	29.28
	Intense dust PV panels	197.83 ± 44.14	0.22	1.53	308.84	75.23	0	0	39.05	60.95
After cleaning	Simple dust PV panels	501.17 ± 94.63	0.19	1.86	5.5	1.75	1	10	98.91	1.09
	Moderate dust PV panels	471 ± 98.71	0.21	1.28	35.67	7.10	2	20	92.96	7.31
	Intense dust PV panels	314.67 ± 92.75	0.29	2.13	192	38.45	2	20	62.11	37.89

Avg.±SD: average ± standard deviation; CV: coefficient of variation; SE: standard error; Δ Output power (W): differences in output power between standard PV panels and other different dust PV panels; ΔE*_{ab}: total color difference; Δ Efficiency (%): differences in efficiency between standard PV panels and other different dust and clean PV panels.

compared to the power produced before the cleaning process. Hence, the simple dust solar panels achieved the highest rate of recovery of lost power in the afternoon from 13 to 14 hours, as the value of the produced power increased from 534 W before cleaning to 603 W after the cleaning process was carried out from one cleaning round only. Solar panels with moderate dust came in the second rank in recovering the lost power because of dust accumulation on them at a lower rate than solar panels with simple dust in the same afternoon period from 13 to 14 hours. The value of the produced power increased from 435 W before cleaning to 581 W

after the cleaning process was carried out from two cleaning rounds only. Thirdly, the solar panels with intense dust occupied the last rank in recovering the lost power as a result of the thick accumulation of dust on the solar PV panel surfaces. Although the cleaning process was carried out in two rounds, the heavy accumulation of dust led to the adhesion of a layer of dust to the solar PV panel surfaces, which caused a decrease in the power produced by the PV panels.

Table 3 illustrates the differences in power produced and color for PV panels before and after cleaning compared to the clean PV solar panels (standard color) and statistical

details (e.g. Avg. $\pm$ SD, CV, and SE), at different stages. In this regard, the intense dust PV panels achieved the lowest value of output power with a general mean value of 197.83 ± 44.14 W/day with CV and SE values of 0.22 and 1.53. This is due to the accumulation of a thick dust layer and showed the highest value of total color differences (ΔE^*_{ab}) of 75.23 compared to the standard color of the clean PV panels. This great value of ΔE^*_{ab} between the clean PV panels and PV panel samples with intense dust led to a dramatic decrease in the efficiency of the PV solar panels. Where the differences in efficiency (%) between standard PV panels and the intense dust PV panels by up to a value of 60.95%. The cleaning robot implemented a quick cleaning process of two cleaning rounds within a time of 10 s/round. As a result of this cleaning process, the solar PV panel efficiency has been raised to a value of 62.11% with Δ an efficiency (%) of 37.89 compared to the full efficiency of the standard PV panel. Perhaps, this efficiency is due to the presence of dust and bird droppings attached to the surfaces of the solar cells, which requires a washing cycle. These results are consistent with [19, 35, 38].

As well, the Δ efficiency value of the PV panels with simple and moderate dust compared to the full efficiency of standard PV panels reached 1.09 and 7.31% at small total color differences' (ΔE^*_{ab}) values of 1.75 and 7.10.

4. Conclusions

This investigation is aimed at providing a practical approach to automate both monitoring and cleaning of the PV panel's surfaces through the design and manufacture dry-cleaning robot based on the dust accumulation monitoring system, using an image processing system and color analysis of the PV panel surfaces. In order to clean the PV panel surfaces regularly and raise the efficiency of PV solar panels to generate electricity, it was observed that there is a significant difference in the total color (ΔE^*_{ab}) between panels with clean surfaces compared to panels with different dust densities (simple, moderate, and intense) with values of 0, 43.69, 61.19, and 75.23, respectively. These color differences because of the differences in dust densities indicated a decrease in the efficiency of electric power production between clean PV panels and other panels with simple, moderate, and intense dust with values of 100, 88.03, 70.72, and 39.05%. Then, the cleaning robot was operated to perform a cleaning process consisting of one cleaning round for simple dust PV panels and two cleaning rounds for moderate and intense PV panels, at a rate of one round/10 seconds. After carrying out the cleaning process, the efficiency of the solar panels for power production increased to reach 98.91, 92.96, and 62.11 for simple, moderate, and intense dust PV panels, respectively. Thus, it can be seen that this robot combined with a color monitoring system will be more effective in solar PV panel systems on a large scale.

5. Future Work

- (i) This robot will be developed in future work with more solid and effective materials suitable for work-

ing in the external environment during the design and manufacturing operations to be more suitable for large-scale solar stations

- (ii) The robots will also be linked in one location to the control unit so that the status of the robot and its efficiency can be monitored, and thus, any problems that hinder its work can be solved
- (iii) Developing a color monitor system for solar panels using automatic vision systems to be a movement mechanism and work according to a schedule to work on a larger scale
- (iv) In addition to designing a more accurate control unit using the programmable logic controller (PLC) and linking it to the Global Positioning System (GPS).

Data Availability

The authors confirm that the data supporting the findings of this study are available within the article. Furthermore, we are fully prepared to cooperate by making the data available upon reasonable request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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